Computing Depth

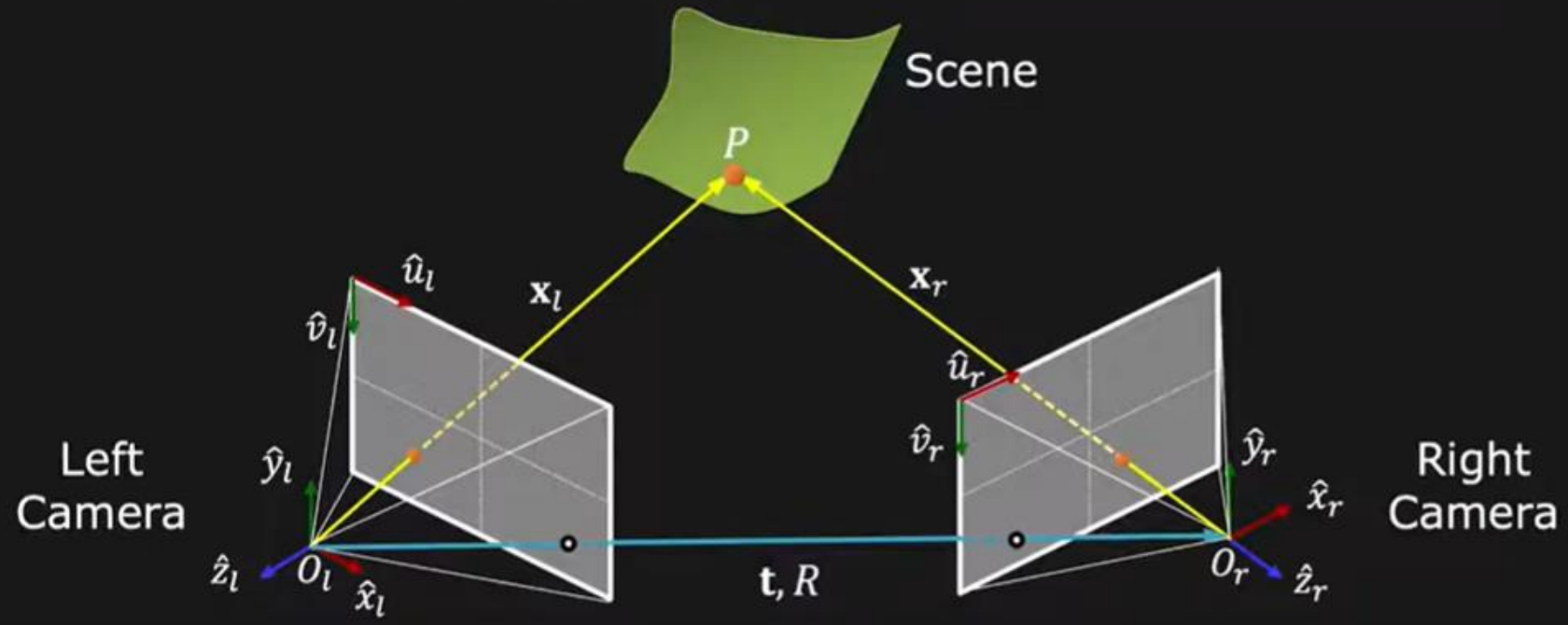
Shree K. Nayar

Columbia University

Topic: Uncalibrated Stereo, Module: Reconstruction II

First Principles of Computer Vision

Computing Depth



Given the intrinsic parameters, the projections of scene point on the two image sensors are:

$$\begin{bmatrix} u_l \\ v_l \\ 1 \end{bmatrix} \equiv \begin{bmatrix} f_x^{(l)} & 0 & o_x^{(l)} & 0 \\ 0 & f_y^{(l)} & o_y^{(l)} & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_l \\ y_l \\ z_l \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} u_r \\ v_r \\ 1 \end{bmatrix} \equiv \begin{bmatrix} f_x^{(r)} & 0 & o_x^{(r)} & 0 \\ 0 & f_y^{(r)} & o_y^{(r)} & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \\ 1 \end{bmatrix}$$

Computing Depth

Left Camera Imaging Equation

$$\begin{bmatrix} u_l \\ v_l \\ 1 \end{bmatrix} \equiv \begin{bmatrix} f_x^{(l)} & 0 & o_x^{(l)} & 0 \\ 0 & f_y^{(l)} & o_y^{(l)} & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_l \\ y_l \\ z_l \\ 1 \end{bmatrix}$$

Right Camera Imaging Equation

$$\begin{bmatrix} u_r \\ v_r \\ 1 \end{bmatrix} \equiv \begin{bmatrix} f_x^{(r)} & 0 & o_x^{(r)} & 0 \\ 0 & f_y^{(r)} & o_y^{(r)} & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \\ 1 \end{bmatrix}$$

We also know the relative position and orientation between the two cameras.

$$\begin{bmatrix} x_l \\ y_l \\ z_l \\ 1 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \\ 1 \end{bmatrix}$$

Computing Depth

Left Camera Imaging Equation:

$$\begin{bmatrix} u_l \\ v_l \\ 1 \end{bmatrix} \equiv \begin{bmatrix} f_x^{(l)} & 0 & o_x^{(l)} & 0 \\ 0 & f_y^{(l)} & o_y^{(l)} & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \\ 1 \end{bmatrix}$$

Right Camera Imaging Equation:

$$\begin{bmatrix} u_r \\ v_r \\ 1 \end{bmatrix} \equiv \begin{bmatrix} f_x^{(r)} & 0 & o_x^{(r)} & 0 \\ 0 & f_y^{(r)} & o_y^{(r)} & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \\ 1 \end{bmatrix}$$

Computing Depth

Left Camera Imaging Equation:

$$\begin{bmatrix} u_l \\ v_l \\ 1 \end{bmatrix} \equiv \begin{bmatrix} f_x^{(l)} & 0 & o_x^{(l)} & 0 \\ 0 & f_y^{(l)} & o_y^{(l)} & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \\ 1 \end{bmatrix}$$

$$\tilde{\mathbf{u}}_l = P_l \, \tilde{\mathbf{x}}_r$$

Right Camera Imaging Equation:

$$\begin{bmatrix} u_r \\ v_r \\ 1 \end{bmatrix} \equiv \begin{bmatrix} f_x^{(r)} & 0 & o_x^{(r)} & 0 \\ 0 & f_y^{(r)} & o_y^{(r)} & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \\ 1 \end{bmatrix}$$

$$\tilde{\mathbf{u}}_r = M_{int_r} \, \tilde{\mathbf{x}}_r$$

Computing Depth

The imaging equations:

$$\tilde{\mathbf{u}}_r = M_r \, \tilde{\mathbf{x}}_r \qquad \tilde{\mathbf{u}}_l = P_l \, \tilde{\mathbf{x}}_r$$

$$\begin{bmatrix} u_r \\ v_r \\ 1 \end{bmatrix} \equiv \begin{bmatrix} m_{11} & m_{12} & m_{13} & m_{14} \\ m_{21} & m_{22} & m_{23} & m_{24} \\ m_{31} & m_{32} & m_{33} & m_{34} \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \\ 1 \end{bmatrix} \qquad \begin{bmatrix} u_l \\ v_l \\ 1 \end{bmatrix} \equiv \begin{bmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \\ 1 \end{bmatrix}$$

Known Unknown Known Unknown

Rearranging the terms:

$$\begin{bmatrix} u_r m_{31} - m_{11} & u_r m_{32} - m_{12} & u_r m_{33} - m_{13} \\ v_r m_{31} - m_{21} & v_r m_{32} - m_{22} & v_r m_{33} - m_{23} \\ u_l p_{31} - p_{11} & u_l p_{32} - p_{12} & u_l p_{33} - p_{13} \\ v_l p_{31} - p_{21} & v_l p_{32} - p_{22} & v_l p_{33} - p_{23} \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \end{bmatrix} = \begin{bmatrix} m_{14} - m_{34} \\ m_{24} - m_{34} \\ p_{14} - p_{34} \\ p_{24} - p_{34} \end{bmatrix}$$

Computing Depth: Least Squares Solution

$$\begin{bmatrix} u_r m_{31} - m_{11} & u_r m_{32} - m_{12} & u_r m_{33} - m_{13} \\ v_r m_{31} - m_{21} & v_r m_{32} - m_{22} & v_r m_{33} - m_{23} \\ u_l p_{31} - p_{11} & u_l p_{32} - p_{12} & u_l p_{33} - p_{13} \\ v_l p_{31} - p_{21} & v_l p_{32} - p_{22} & v_l p_{33} - p_{23} \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \end{bmatrix} = \begin{bmatrix} m_{14} - m_{34} \\ m_{24} - m_{34} \\ p_{14} - p_{34} \\ p_{24} - p_{34} \end{bmatrix}$$

$A_{4\times3}$ (Known) $\quad \mathbf{x_r}$ (Unknown) $\quad \mathbf{b}_{4\times1}$ (Known)

Find least squares solution using pseudo-inverse:

$$A\mathbf{x_r} = \mathbf{b}$$

$$A^T A\mathbf{x_r} = A^T\mathbf{b}$$

$$\boxed{\mathbf{x_r} = (A^T A)^{-1} A^T \mathbf{b}}$$

3D Reconstruction with Internet Images

St. Peter's Basilica (1275 Images)



[Snavely 2006]

20 Shree K. Nayar

3D Reconstruction with Internet Images

St. Peter's Basilica (1275 Images)



[Snavely 2006]

3D Reconstruction with Internet Images

Piazza San Marco (13709 Images)



[Furukawa 2010]

020 Shree K. Nayar

Active Stereo Results



Left Image

Right Image



3D Structure

[Zhang 2003]

©20 Shree K. Nayar

Active Stereo Vision

Created in August 15, 2024

2024 · #Active #Stereo #Vision · 3D surface modeling

Active Stereo Vision is a computer vision technique used to generate 3D depth maps of objects and scenes. It is based on the principles of stereo vision and enhances the accuracy and reliability of depth perception by actively projecting structured light or infrared light.

Principles of Active Stereo Vision

1. **Basic Stereo Vision**:
    - Stereo vision simulates the principle of human binocular disparity. Two cameras (commonly referred to as the left and right cameras) capture images of the same scene simultaneously. Due to the different positions of the two cameras, the same object appears slightly shifted in the two images; this shift is known as disparity.
    - By comparing corresponding points in the two images (e.g., edges or corners of objects), the disparity value for each pixel can be calculated. A larger disparity indicates that the object is closer to the cameras, while a smaller disparity indicates that the object is farther away.
    - Using triangulation principles and the known baseline distance between the cameras (i.e., the distance between the two cameras), the actual depth of the object relative to the cameras can be computed.
2. **Enhanced by Active Projection**:
    - In traditional stereo vision systems, if the scene being captured lacks texture (such as smooth surfaces or uniform colors), it becomes difficult for the cameras to find enough feature points for matching. This can reduce the accuracy of depth perception.
    - To enhance the accuracy of stereo vision systems, active stereo vision uses a structured light or infrared projector to cast a known pattern (such as a dot matrix or stripes) onto the scene. This pattern creates artificial textures on the surface, making it easier for the cameras to find and match feature points in the images.
    - The projected light is usually infrared, as it is outside the visible light spectrum and does not interfere with human vision or visual images.
3. **Depth Calculation**:
    - The cameras capture the deformed pattern projected onto the scene. By analyzing these deformed patterns, stereo vision algorithms can more accurately calculate the disparity and, consequently, determine the depth information of each point in the scene.

Application Scenarios

Active stereo vision systems are commonly used in various applications, such as:

- **Robotic Navigation**: Provides real-time 3D environmental perception, helping robots avoid obstacles and plan paths.
- **Gesture Recognition**: Accurately captures the 3D position and movement trajectory of hands, enabling natural human-computer interaction.
- **Augmented Reality**: Supplies precise 3D scene data for virtual objects, enabling more realistic overlay effects.
- **3D Scanning**: Used to generate high-precision 3D models of objects, widely applied in medical, industrial design, and other fields.

Summary

Active stereo vision leverages the disparity principle of stereo vision and enhances depth perception by actively projecting known patterns. This technology is highly valuable in computer vision, particularly in scenarios requiring high accuracy.